Correlation reductions and the exact square-torus parameters

Human Compression project

Abstract

We supply the definite diagram-(4) reduction stated on physical page 9 of Fernandes–Leemans–Weiss, and the definite presentation statements accompanying the experiments concerning diagram (5). Explicit inverse presentation maps give the parameter transformations. For every existing regular fork hypertope, its correlation extension has the intersection property and is the stated universal polytope. The group identities do not assert existence of a hypertope for every period or degenerate quotient.

1 The literal groups and their correlation

Write $a = \rho_0$, $b = \rho_1$, $c = \rho_2$, $d = \rho_3$, with b the central node of the fork. Its Coxeter relations are

$$a^2 = b^2 = c^2 = d^2 = (ab)^4 = (bc)^4 = (bd)^3 = (ac)^2 = (ad)^2 = (cd)^2 = 1. \tag{1}$$

Let H_s^0 be this presented group with $(abcb)^s = 1$, and let H_s^1 have instead $(abc)^{2s} = 1$, where s is a positive integer. These are the source's axial $(s, 0)$ and diagonal (s, s) square-torus residue relations, respectively. The assertion here concerns these entire presentations, not a finite image of them.

The substitution $\alpha : a \leftrightarrow c$, fixing b, d, preserves (1). Put $R = abcb$ and $Q = abc$. Then

$$\alpha(R) = cbab = bR^{-1}b, \qquad \alpha(Q) = cba = Q^{-1}.$$

It therefore preserves each of the two additional normal closures. It induces an automorphism of H_s^ϵ with square one, whether or not that presented group already defines a nondegenerate geometry. Form the external semidirect product

$$E_s^\epsilon = H_s^\epsilon \rtimes_\alpha \langle t \rangle, \qquad t^2 = 1.$$

Thus $tat = c$, $tbt = b$, $tdt = d$. The external factor has order two even in a collapsed small-parameter group. All statements about actual hypertope types below impose their stated nondegeneracy hypotheses.

Theorem 1 (Exact universal presentation reduction)**.** *On the ordered generators (t, a, b, d), E_s^0 is the Coxeter group $[4, 4, 3]$ with the extra relation $(tab)^{2s} = 1$, and E_s^1 is $[4, 4, 3]$ with the extra relation $(taba)^{2s} = 1$. In particular their marked local square-torus presentations are respectively (s, s) and $(2s, 0)$. In both cases H_s^ϵ is exactly the kernel of the character which sends t to -1 and a, b, d to 1.*

Proof. Eliminate c using $c = tat$. The relation $(ac)^2 = 1$ becomes $(at)^4 = 1$. The relations involving bc and cd are conjugates by t of those involving ba and ad; t commutes with b, d. The remaining relations are exactly the string $[4, 4, 3]$ relations on (t, a, b, d). For the periods the following are word identities, not order tests:

$$abcb = abtatb = (abt)^2 = t(tab)^2t, \tag{2}$$

$$(abc)^2 = abacbc = (abat)^2 = (taba)^{-2}. \tag{3}$$

In the first line commuting t and b gives $(abt)^2$. In the second line commute a, c, then use $tat = c$ and $tbt = b$. Taking the indicated powers proves equivalence of the extra relators and their full normal closures.

Conversely start with either stated string presentation, set $c = tat$, and retain t. The preceding steps give every fork relation, the correct period, and the conjugation rules for α. The resulting homomorphisms between that presentation and E_s^ϵ are inverse on all generators. This proves isomorphism of the presented groups, including the universal upper direction. Every defining string relator has even t-parity, so the character exists and is onto. In the semidirect product its kernel is precisely H_s^ϵ. Equivalently, the Schreier transversal $\{1, t\}$ gives generators a, c, b, d and the already displayed fork presentation; no omitted conjugate relator is required. □

The usual square-torus convention for a string marking (x, y, z) is $(xyzy)^q = 1$ for $(q, 0)$ and $(xyz)^{2q} = 1$ for (q, q). Thus (2) and (3), with $(x, y, z) = (t, a, b)$, give exactly the two parameter transformations in the theorem. This uses the same torus convention as the source and the retained square theory; it does not identify a torus by its group order. The convention can also be checked on the square plane: take

$$a(x, y) = (-x, y), \quad c(x, y) = (x, -y), \quad b(x, y) = (1 - y, 1 - x), \quad t(x, y) = (y, x).$$

Here $abcb$ translates by $(-2, 0)$, $(abc)^2$ by $(-2, 2)$, and $taba$ by $(1, -1)$. Rotation-conjugate period vectors give the complete two-dimensional period lattices. The string grid has basis $(1, -1), (1, 1)$, so its diagonal (s, s) lattice is the fork's axial $2s\mathbb{Z}^2$, while its axial $(2s, 0)$ lattice is generated by $(2s, -2s), (2s, 2s)$. This confirms both the scaling and the exchange of axial and diagonal labels. These coordinates are only the local square-plane model, not a purported model of the rank-four group.

2 Intersection property and the geometric assertion

A set of involutions has the intersection property if its generated subgroups satisfy $H_I \cap H_J = H_{I \cap J}$ for all subsets of its marking. The following elementary form of the usual parabolic patching argument suffices; no geometry is inferred just from an abstract group isomorphism. We retain the standard equivalence between string C-groups and regular abstract polytopes, and the regular-hypertope maximal-parabolic coset representation, as explicit foundational inputs.

Lemma 1 (Parabolic patching)**.** *If every maximal proper generated parabolic has the intersection property, and every two distinct such parabolics intersect in their common generated subgroup, then the entire marking has the intersection property.*

Proof. Let G_I, G_J be proper generated subgroups. Choose a missing marker i of I and a missing marker j of J. If $i = j$, the intersection property in that maximal parabolic applies. Otherwise an element of their intersection lies in the common parabolic G_T, where T omits i, j. Applying the intersection property first inside $G_{\bar{i}}$ and then inside $G_{\bar{j}}$ puts it in $G_{I \cap T}$ and $G_{J \cap T}$. Both are in G_T, which inherits the intersection property from a maximal parabolic. Their intersection is $G_{I \cap J}$, as required. Cases with I or J the full marking are immediate. □

Theorem 2 (From each existing fork to its polytope)**.** *Suppose $(H_s^\epsilon; a, b, c, d)$ is the actual marked group of a regular hypertope of source diagram (4), with the prescribed square-torus residue and the prescribed cubical residues. Then $(E_s^\epsilon; t, a, b, d)$ is a string C-group. Its polytope is respectively*

$$\big\{\{4, 4\}_{(s,s)}, \{4, 3\}\big\}, \qquad \big\{\{4, 4\}_{(2s,0)}, \{4, 3\}\big\},$$

and is universal with those marked local defining presentations. The given fork hypertope is recovered from its parity kernel and marking.

Proof. The marked group of the given regular hypertope has the intersection property. In $E = E_s^\epsilon$ put $L = \langle t, a, b\rangle$ and $K = \langle a, b, d\rangle$. The semidirect normal form gives

$$L \cap H_s^\epsilon = \langle a, b, c\rangle, \qquad \langle t, a\rangle \cap H_s^\epsilon = \langle a, c\rangle.$$

Indeed the listed kernel generators and t generate the respective subgroups, and t normalizes each listed kernel. Consequently

$$\langle t, a\rangle \cap \langle a, b\rangle = \langle a, c\rangle \cap \langle a, b\rangle = \langle a\rangle.$$

The two involutions a, c are independent commuting markers, so ta has order four. The remaining maximal subgroup of (t, a, b) is $\langle t, b\rangle = \langle t\rangle \times \langle b\rangle$. Its intersections with $\langle t, a\rangle$ and $\langle a, b\rangle$ are respectively $\langle t\rangle$ and $\langle b\rangle$, using parity and $\langle a, c\rangle \cap \langle b\rangle = 1$. Each two-marker subgroup has the intersection property, so Lemma 1 makes (t, a, b) a C-group. The other end deletion (a, b, d) inherits the intersection property and its string diagram directly from H_s^ϵ. Finally

$$L \cap K = \langle a, b, c\rangle \cap \langle a, b, d\rangle = \langle a, b\rangle.$$

For completeness the other two maximal parabolics are $M_1 = \langle t, b, d\rangle = \langle t\rangle \times \langle b, d\rangle$ and $M_2 = \langle t, a, d\rangle = \langle t, a\rangle \times \langle d\rangle$. The first product is direct by parity; the second uses $\langle a, c\rangle \cap \langle d\rangle = 1$ and the displayed commutations. Thus they too have the intersection property. In the ordering $(M_0, M_1, M_2, M_3) = (K, M_1, M_2, L)$ the six intersections are

(i, j)	$(0,1)$	$(0,2)$	$(0,3)$	$(1,2)$	$(1,3)$	$(2,3)$
$M_i \cap M_j$	$\langle b, d\rangle$	$\langle a, d\rangle$	$\langle a, b\rangle$	$\langle t, d\rangle$	$\langle t, b\rangle$	$\langle t, a\rangle$.

To verify the last three, both groups contain t, so intersect their parity kernels $\langle b, d\rangle$, $\langle a, c, d\rangle$, $\langle a, b, c\rangle$ inside H and then reattach $\langle t\rangle$. The first three follow in the same way with $K \subset H$. The original intersection property supplies every kernel intersection. Lemma 1 proves the full string intersection property. Its cubical vertex figure is the given (a, b, d) residue. Its facet group is the correlation extension of the given entire (a, b, c) torus group. Applying the rank-three presentation calculation of Theorem 1 identifies the facet with the stated torus, without an additional local quotient or period collapse.

Theorem 1 supplies exactly the global string relations and the defining facet period, with no further relators. Hence every regular polytope of those marked local types is a quotient, and the displayed polytope is universal. Conversely its exact parity kernel is the original marked group. A regular hypertope with that marking is its standard maximal-parabolic coset geometry, so this recovers the original geometry, not just an unmarked group order. □

The type exchange $a \leftrightarrow c$ is an actual correlation of the given regular fork geometry: on its maximal-parabolic cosets apply α and exchange the corresponding two types. This preserves coset intersection and is an involution. The new rank-four generating list is t, a, b, d. The source's three-letter parenthesis omits ρ_3; t, a, b generates the facet extension, while the complete rank-four assertion requires $d = \rho_3$ as well.

3 Diagram (5): definite algebra, separate geometric experiment

For diagram (5), keep the labels a, b, c of its square-torus path, with b central, and put d at the other vertex of the diamond. Its Coxeter relations are

$$a^2 = b^2 = c^2 = d^2 = (ab)^4 = (bc)^4 = (ad)^3 = (cd)^3 = (ac)^2 = (bd)^2 = 1. \tag{4}$$

This agrees with the actual source drawing: the two edges at b have label four and the two at d have the undisplayed label three. Let D_s^0, D_s^1 add, respectively, $R^s = 1$ and $Q^{2s} = 1$ to (4). The same substitution $a \leftrightarrow c$, fixing b, d, is an involutory automorphism of both groups.

Theorem 3 (Diamond-to-fork presentations)**.** *The extension $D_s^0 \rtimes \langle t \rangle$, marked (t, a, b, d) with a central, has exactly the diagram-(4) fork presentation with torus parameter (s, s). The extension $D_s^1 \rtimes \langle t \rangle$ has exactly that fork presentation with torus parameter $(2s, 0)$. In both cases D_s^ϵ is its exact t-parity kernel. These statements hold for all positive integers s as isomorphisms of the displayed presented groups.*

Proof. Again eliminate $c = tat$. The relation $(ac)^2$ gives $(ta)^4$; $(cd)^3$ is the t-conjugate of $(ad)^3$; $(bc)^4$ is the conjugate of $(ba)^4$. The remaining Coxeter relations give the fork with centre a, arms t, b of label four and arm d of label three. The period words transform by exactly (2)–(3). Thus they become $(tab)^{2s}$ and $(taba)^{2s}$, respectively. Now t, a, b is the fork's toroidal path, so these are its diagonal (s, s) and axial $(2s, 0)$ periods. The inverse substitution gives all original relations and the conjugation rules, proving both directions and the exact kernel assertion. In particular the source's conjugated word is correct: $\alpha(R) = cbab = (cbt)^2$ and $cbt = tab$. Conjugating $R^s = 1$ therefore gives precisely $(cbt)^{2s} = 1$. □

If the original diamond marking is a C-group, the extended fork marking also has the intersection property. The six maximal-parabolic intersections in the proof of Theorem 2 are unchanged. The one modification is $M_2 = \langle t, a, d \rangle$: it is now a rank-three string marking of type $\{4, 3\}$, rather than a direct product. Its three two-marker intersections follow from $\langle t, a \rangle \cap H = \langle a, c \rangle$, the original intersection property and $[t, d] = 1$, exactly as for $\langle t, a, b \rangle$. Lemma 1 applies first there and then to the full marking. This still does not prove flag transitivity of the new nonstring coset incidence system. The source explicitly calls hypertope production through this correlation experimental; the definite group and period assertions above do not remove that geometric qualification.

4 Exact scope and converse limitations

The group isomorphisms hold for every positive integer s. The geometric determination is quantified over each existing regular universal fork hypertope with its stated actual residues. It determines its marked group and coset geometry from the corresponding universal polytope. It does not claim that every period in the displayed presentations gives a hypertope, or that every quotient preserves its local types, intersection property or flag transitivity. In a proposed converse construction those properties must be established, not deduced merely from the parity map. For example, the separately proved bipartite halving theorem applies to a regular lattice polytope with a bipartite vertex graph; its actual lattice and incidence hypotheses must be checked before using it. No such unchecked hypothesis is inserted into Theorem 2.

This repairs the earlier combined documentary disposition for diagrams (4) and (5). Diagram (4) has the definite universal reduction just proved. Diagram (5)'s exact stated presentation and parameter transformations are proved separately in Theorem 3; its experimental hypertope-existence wording remains explicitly qualified. Neither an unproved flag-transitivity claim nor the order-768 future-work example is silently added to this supplement.

The source is M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, the exact retained 21-page 2018 edition, physical pages 9–10. The correlation idea and parameter reduction are theirs. The work here writes the inverse presentation maps and the missing marked-intersection argument, reusing the established Schreier and string-C-group principles. It claims no new original admission, historical priority, question solution or Lean formalization. No presentation enumeration or previously accepted geometric census is repeated.